

JBM9204 INTRODUCTION TO BIOMETRIC SYSTEMS



UNIT-2 FINGERPRINT IDENTIFICATION TECHNOLOGY

CLASS – III AI & ML

SEMESTER – V

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Unit-II FINGERPRINT IDENTIFICATION TECHNOLOGY

COURSE OBJECTIVE: To emphasize concepts and design of fingerprint recognition biometric systems.

SYLLABUS

Biometrics using Fingerprint - General description of fingerprints - Fingerprint Recognition system - fingerprint sensors using RF imaging techniques - Finger print feature processing techniques - Minutiae Extraction – Computer enhancement and modelling of fingerprint images - fingerprint classification – fingerprint matching.

COURSE OUTCOME 2: Extract the fingerprint features, and apply for fingerprint recognition.

Biometrics using Fingerprint

- ▶ **Fingerprint biometrics** → the systematic study and application of unique physical attributes inherent in an individual's fingerprints.
- ▶ Represents a more dependable identification method than traditional passwords or identity cards, fingerprint biometrics eliminates the issues of misplacement, forgetfulness, or theft.
- ▶ The distinctive nature of each person's fingerprint ensures a robust barrier against unauthorized access to secure data.

Biometrics using Fingerprint

- ▶ **A fingerprint** → the representation of the epidermis of a finger → consists of a pattern of interleaved ridges and valleys.
- ▶ Fingerprints can have an arching, looping, or swirling pattern, but the exact patterns differ from person to person.
- ▶ Fingerprint ridges form through a combination of genetic and environmental factors.
- ▶ Even identical twins have different fingerprints.
- ▶ Formed (i.e. became stable) at about seven months of fetus development and finger ridge configurations do not change throughout the life of an individual, except in case of accidents such as cuts on the fingertips → This property makes fingerprints a very attractive biometric identifier.

Biometrics using Fingerprint

Why do we have fingerprint?

1. **Improved Grip** → The raised ridges increase friction between our skin and objects, helping us hold things securely.
2. **Enhanced Touch Sensitivity** → The ridges amplify vibrations when our fingers interact with surfaces, sending more signals to the brain.
3. **Protection and Adaptation** → The ridges help channel away water or sweat, maintaining a secure grip even in damp conditions.



Biometrics using Fingerprint

History of Fingerprint Recognition

Fingerprints have been found on ancient archaeological artifacts, showing that people recognized their uniqueness long ago.

Sl. No.	Year	Involved Scientist	Discovery / Proposal
1.	1686	Marcello Malpighi	Described fingerprint patterns such as ridges, loops, and spirals
2.	1880	Henry Faulds	Proposed that fingerprints are unique to each individual.
3.		Sir William Herschel	Demonstrated the practical use of fingerprints for personal identification
4.	1888	Sir Francis Galton	Introduced the concept of minutiae features for fingerprint matching

Biometrics using Fingerprint

History of Fingerprint Recognition

Sl. No.	Year	Involved Scientist / Org.	Discovery / Proposal
1.	1899	Sir Edward Henry	Developed a classification system- became the standard method for organizing fingerprints
2.	Early 20th century		Fingerprint recognition was officially adopted for criminal identification and forensic investigations
3.	1924	FBI	Established its fingerprint identification division and began maintaining a large fingerprint database
4.	1960's		Automatic Fingerprint Identification Systems (AFIS)

- ▶ Today, AFIS is widely used by law enforcement agencies.
- ▶ Fingerprint recognition has expanded beyond forensics into PC login, smartphones, ATMs, access control, and e-commerce due to its high accuracy and low cost.

Biometrics using Fingerprint

Why fingerprints are unique?

- ▶ Fingerprint pattern formation consists of two components:
 - Developmental and
 - Genetic
- ▶ The combination of genetic blueprint + random physical conditions creates a one-of-a-kind pattern for every individual.
- ▶ Even identical twins (with the same DNA) have different fingerprints because the tiny, random environmental factors in the womb - like pressure, growth stresses, and exact timing of skin development - cannot be replicated.

General Description of Fingerprint

Aspects of fingerprints that are analyzed include:

- ▶ **Patterns:** The general pattern or type of fingerprint (arch, loop, or whorl) is inherited through genetics.
- ▶ **Minutiae:** The precise details of the ridges, known as minutiae, are influenced by random and unpredictable factors such as pressure, blood flow, and position in the womb during development.
- ▶ **Ridges:** Each ridge in a fingerprint contains several minutiae points, which can be bifurcations (where one ridge splits into two) or ridge endings.

Principles of Fingerprints

Fingerprints follow three basic principles

1. **Pattern Type:** All fingerprints can be classified into pattern types.
2. **Uniqueness:** Each fingerprint is unique, no two people have the same fingerprints. Each person's fingerprints differ from finger to finger.
3. **Permanence:** Fingerprints are unchanging from 3 months fetal life until decomposition. Scars are the only exemption to this rule.

General Description of Fingerprint

Structure of the Fingerprint

- ▶ A fingerprint is made up of friction ridges, the raised skin patterns on the fingertips, palms, and soles.
- ▶ A visual inspection of friction skin reveals a series of lines corresponding to hills (ridges) and valleys (grooves).



General Description of Fingerprint

Ridges:

- ▶ The raised, dark lines on the skin of the fingertip.
- ▶ Formed by the upper layer of the epidermis and help with grip and touch sensitivity.
- ▶ Create the unique fingerprint patterns—arches, loops, and whorls—used for identification.
- ▶ Each ridge has sweat pores that leave impressions (latent fingerprints) on surfaces.

Valleys:

- ▶ The lighter, lower spaces between the ridges.
- ▶ Valleys separate the ridges and give fingerprints their distinctive patterns.
- ▶ Play a role in channeling sweat away from the skin surface, maintaining grip.

General Description of Fingerprint

Ridge characteristics (or) Minutiae

- ▶ The fine details in the patterns of friction ridges found on the skin's surface, primarily on the fingers, palms, and soles.
- ▶ Form the foundation of fingerprint analysis in forensic science.
- ▶ The unique ridge features details - such as ridge endings, bifurcations, lakes, and cores are crucial in forensic fingerprint identification.
- ▶ No two fingerprints share the same arrangement of minutiae, making them a highly reliable method for personal identification.
- ▶ The arrangement of these minutiae within a fingerprint is what makes it unique to each individual.
- ▶ **Forensic** experts and automated systems like AFIS (Automated Fingerprint Identification System) use these minutiae to match fingerprints for crime scene investigation, personal identification, and legal proceedings.

General Description of Fingerprint

Ridge Characteristics

Minutiae Type	Description
Crossover	A point where two ridges cross each other.
Core	The center or focal point of the fingerprint pattern, usually in a loop or whorl (Innermost loop of the fingerprint).
Bifurcation or Fork	A point where a single ridge splits into two ridges.
Ridge Ending	The point where a ridge abruptly stops.
Island	A small, isolated ridge or a very short ridge segment.(Very short ridge that stands alone)
Delta	A triangular region where ridge patterns diverge; usually found in loops and whorls.
Pore	Tiny openings on the ridges where sweat is released; visible in high-resolution fingerprint images.

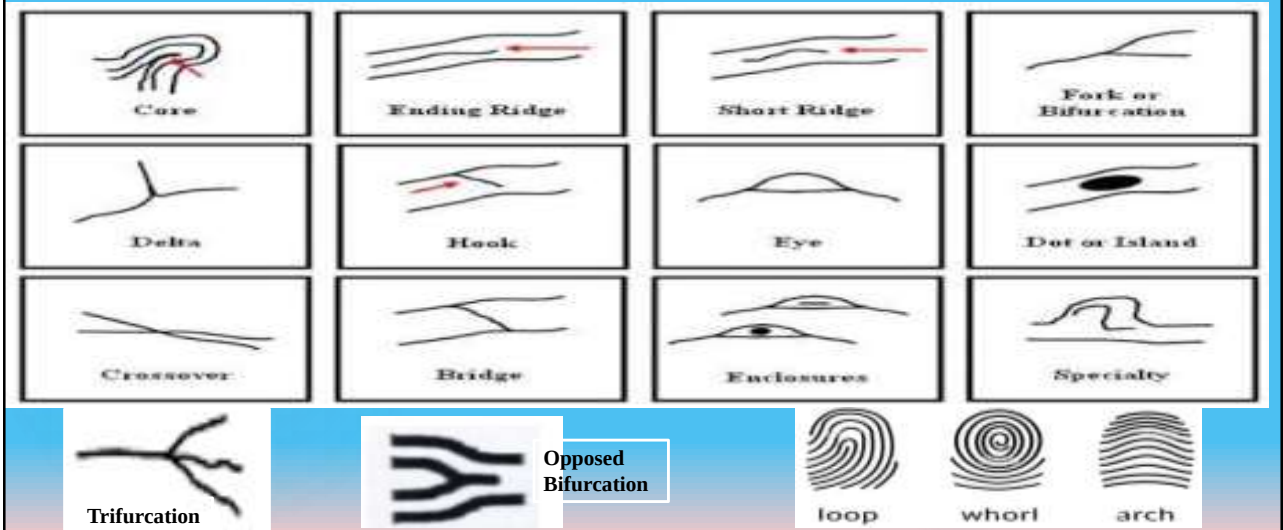
General Description of Fingerprint

Ridge Characteristics

Minutiae Type	Description
Hook (Spur)	A small ridge extending from a longer ridge but not branching off completely.
Trifurcation	A ridge that splits into three distinct ridges, forming a "trident" or fork-like shape.
Opposed Bifurcation	Two bifurcations facing each other where ridges split in opposite directions.
Incipient Ridge	A faint, underdeveloped ridge, shorter or thinner than surrounding ridges.
Bridge	A short friction ridge that runs between and connects two otherwise parallel ridges.

General Description of Fingerprint

Various Minutiae - specific ridge characteristics used in forensic fingerprint identification



General Description of Fingerprint

Fingerprint Ridge Patterns

- ▶ Three main types of fingerprint patterns with several subgroups of patterns:
 - Arch (or curve) pattern
 - Loop pattern
 - Whorls pattern

Arch Patterns

- ▶ Account for approximately 5-15 percent of fingerprint patterns.
- ▶ Categorized into two subgroups: 1. Plain arches, 2. Tented arches

General Description of Fingerprint

1. Plain Arches

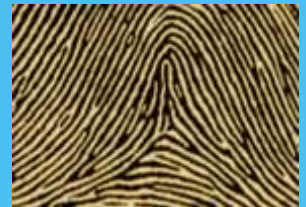
- It consists of ridges that flow from one side of the finger to the other in a smooth, wave-like manner.
- Lacks the complexities of loops or whorls and has no deltas.
- The ridges rise slightly at the center but form no sharp angles or curves.
- Least common fingerprint type, appearing in only about 5% of the population.
- Their simplicity makes them easy to identify.



General Description of Fingerprint

2. Tented Arches

- The ridges rise sharply at the center, creating a peak or tent-like structure.
- Have a distinct central ridge that forms an angle or spike → this makes them more complex.
- The tented arch's unique structure makes it a fascinating pattern in fingerprint analysis. Its sharp rise sets it apart from other types, making it easily distinguishable.
- More common than plain arches, but still relatively rare.
- Tented arches account for about 1-2% of all fingerprints.



General Description of Fingerprint

Loop Patterns

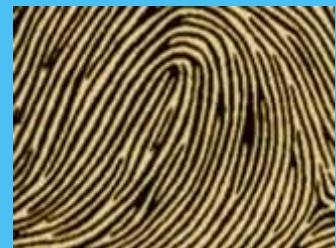
- ▶ Account for approximately 60-65 percent of fingerprint patterns.
- ▶ Most common pattern.
- ▶ In fingerprint classification, the direction of the loop is critical for accurate identification.
- ▶ Categorized into two subgroups based on right slant or left slant
 - 1. Ulnar (ridges flow toward the little finger)
 - 2. Radial (ridges flow toward the thumb)



General Description of Fingerprint

1. Ulnar (ridges flow toward the little finger)

- The most common fingerprint pattern, present in about 60% of individuals.
- Serve as a foundational pattern in the classification and identification of fingerprints.
- These loops curve inward toward the little finger, following the direction of the ulna bone in the forearm.
- The ridges enter and exit on the same side of the finger, forming a loop.
- Easily recognized and widely studied due to their high occurrence rate.



General Description of Fingerprint

2. Radial (ridges flow toward the thumb)

- Fingerprint patterns that curve toward the thumb
- Named after the radius bone, located on the thumb side of the forearm.
- The ridges enter and exit on the same side of the finger, forming a loop that flows in the direction of the thumb.
- Less common than ulnar loops and are found in a smaller percentage of the population.
- Unique due to their specific directional flow, which distinguishes them from other loop patterns.
- Radial loops highlight the importance of understanding ridge flow when analyzing prints.



General Description of Fingerprint

Whorl Patterns

- ▶ Account for 30-35 percent of fingerprint patterns.
- ▶ Categorized into four subgroup:
 - 1. Plain
 - 2. Double loop or Lateral loop
 - 3. Central pocket loop
 - 4. Accidental



General Description of Fingerprint

1. Plain Whorls

- Spiral ridge patterns that form complete loops.
- One of the most easily recognizable fingerprint types.
- Appear in about 30% of the population, making them a relatively common pattern.
- The defining feature of a whorl → its central ridge, which completes at least one full circle.
- Whorls often include two deltas, located on opposite sides of the circular ridge pattern.
- Their clear, circular design makes them easy to classify and compare



General Description of Fingerprint

2. Double loop (or) lateral loops

- Intricate patterns that feature two separate loops intertwining around each other.
- One of the more complex fingerprint types.
- The ridges in a double loop form two distinct centers, creating a pattern that resembles an 'S' or two opposing loops.
- Appear in about 10% of the population, making it a less common but complex pattern.
- Very interesting feature in forensic science.



General Description of Fingerprint

3. Central pocket loop

- Central pocket loops combine elements of loops and whorls. The pattern features a loop with a circular or spiral whorl at the center.
- This pattern is less common than basic loops but more intricate.
- The ridges in a central pocket loop form a distinctive circular shape within the loop, adding complexity to the overall design.



General Description of Fingerprint

4. Accidental

- Accidental fingerprints are the rarest and most complex type.
- They combine features of two or more patterns, often including elements of loops, whorls, and arches. The irregularity of accidental fingerprints makes them challenging to classify.
- They don't fit neatly into any single category, which is why they are labeled as 'accidental'.
- Their unique structure makes them highly distinctive, providing a reliable basis for individual identification.



Fingerprint Recognition system

- ▶ **Fingerprint recognition** → a biometrics-based pattern recognition system widely used to identify or authenticate a person by determining the authenticity of the fingerprint, using an impression or mark made on a surface by one finger.
- ▶ Done manually by a human expert or automatically by a machine.
- ▶ Feature-based (as opposed to image-based) and the features used have a physical interpretation.
- ▶ The terms feature-based and image based methods → indicate the methods used for representing and matching images such as fingerprints.
- ▶ A **feature-based method**, extracts explicit features from the image and encodes these features into a feature set, which is subsequently used for matching.
- ▶ An **image-based method**, directly uses the image for matching without explicitly extracting any features from it.

Fingerprint Recognition system

- ▶ A fingerprint-based biometric system is called either a verification system or an identification system, depending on the context.
- ▶ A **verification system** authenticates one's identity by comparing the fingerprints to a pre-stored biometric template(s) → 1:1
 - It conducts a one-to-one comparison to determine whether the identity claimed by the individual is true.
- ▶ An **identification system** recognizes an individual by searching the entire template database for a match. → 1 : N
 - It conducts one-to-many comparisons to establish the identity of the individual

Fingerprint Recognition system

Fingerprint Acquisition

- ▶ Initially, fingerprints were captured using the **ink-and-paper method** (off-line sensing).
- ▶ The inked fingerprint was scanned to create a digital image.
- ▶ Latent fingerprints collected from crime scenes are also examples of off-line acquisition.
- ▶ Modern systems use live-scan fingerprint scanners, where users place their finger directly on the scanner surface without using ink.
- ▶ Live-scan scanners are widely used in civil and criminal AFIS.

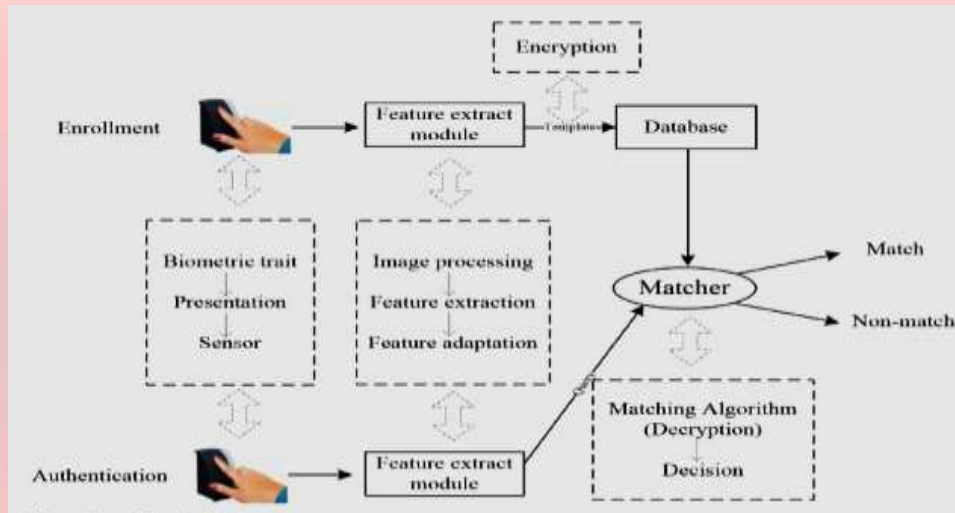


Fingerprint Recognition system (FRS)

- ▶ A typical biometric system consists of four modules: A sensor module, a feature extraction module, a template database, and a matching module.
- ▶ FRS Carries out authentication in two stages: Enrollment and verification.
- ▶ In **enrollment**, a user presents the fingerprint image to be acquired by the sensor module.
 - ▶ Certain features are extracted, and transformed to generate template data for comparison in the next verification stage.
- ▶ In **verification**, the fingerprint image of a query is collected by the sensor module and then compared with the existing template data. The process:
 - ▶ Acquisition of a fingerprint as a **query**.
 - ▶ Preprocessing → simplifies the input fingerprint for feature extraction.
 - ▶ Feature extraction → the fingerprint is processed to generate discriminative properties, called feature vectors.
 - ▶ Matching → the feature vector is compared against existing templates stored in a database.

Fingerprint Recognition system

Fingerprint Recognition System



Fingerprint Recognition system (FRS)

Various types of fingerprint matching techniques

- ▶ Responsible for the recognition of fingerprint patterns.
- ▶ A better matching technique - provides a better result during matching.
- ▶ Some of the fingerprint matching techniques and identification keys are:
 - ▶ Minutiae based
 - ▶ Pattern based
 - ▶ Feature based
 - ▶ ROI (Region of Interest) based
 - ▶ Correlation based
 - ▶ Statistics based
 - ▶ ONNC (Optical Neural Network Computer)

Fingerprint Recognition system (FRS)

- ▶ Fingerprint recognition accuracy depends on:
 - Scanner quality.
 - Sensing area.
 - Image resolution.
 - ▶ Poor-quality scanners reduce recognition performance
- The main parameters characterizing a digital fingerprint**

▶ Resolution

- ▶ Indicates the number of dots or pixels per inch (dpi). Though 500 dpi is the minimum resolution for FBI-compliant scanners, 250 to 300 dpi is the minimum resolution needed for extraction algorithms to locate the minutiae in fingerprint patterns.

▶ Area

- ▶ The size of the rectangular area sensed by a fingerprint scanner. The bigger the area, the more ridges, and valleys are captured, and the more the fingerprint becomes distinctive.

Fingerprint Recognition system (FRS)

The main parameters characterizing a digital fingerprint

▶ Dynamic range (or depth):

- ▶ Denotes the number of bits used to encode each pixel's intensity value. The FBI standard for depth of the pixel bit is 8 bits, yielding 256 gray levels.

▶ Geometric accuracy:

- ▶ It is specified by the maximum geometric distortion introduced by the acquisition device, and expressed as a percentage concerning x and y directions.

▶ Image quality:

- ▶ It is not easy to precisely define the quality of a fingerprint image. Image quality is key to identify the intrinsic finger status. If the fingers are too moist or too dry, or when they are incorrectly presented, most of the scanners produce poor quality images.

Fingerprint Recognition system (FRS)

Benefits of FRS

- ▶ High uniqueness, accuracy, performance, and stability.
- ▶ Easy to use and reliable than other techniques.
- ▶ Good performance, low cost, and small size.
- ▶ Have approximately 50% popularity in the market as compared to other techniques.
- ▶ Has an irreplaceable role of latent prints in crime investigation.
- ▶ Enable government systems to use large databases of fingerprints with a criminal history.

Key challenges in FRS

- ▶ Mismatching of fingerprints due to physical distortions, finger injuries and cuts.
- ▶ Vanishing or fading fingerprint due to jobs like cleaning.
- ▶ Mismatching due to displacement or rotation while scanning the finger.
- ▶ Unauthorized access due to finger plasticity or clay printing.
- ▶ Variability between impressions due to skin conditions, noise in the sensor.
- ▶ Small area fingerprint sensors, providing less information.
- ▶ Fake fingerprints i.e., clays or dummy printing.

Fingerprint Sensors using RF imaging Techniques

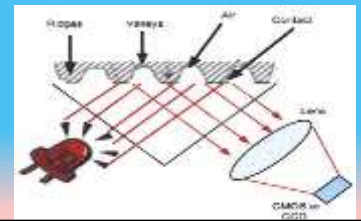
- ▶ Fingerprint sensing → one of the most widely deployed techniques in biometric sensing using sensor technology.
- ▶ Fingerprint sensors can be categorized by the way the user interacts with them, such as contact, contactless, slap, partial, etc.
- ▶ Several sensing mechanisms have been used to detect the finger's ridge-valley structure.
- ▶ **Types of sensors:**
 - a. Optical sensors
 - b. Solid state sensors
 - c. Capacitive sensors → RF Sensor
 - d. Ultrasonic sensors → RF Sensor
 - e. Thermal sensors

Fingerprint Sensors using RF imaging Techniques

Types of Fingerprint Sensors

a) Optical Sensors

- ▶ Frustrated Total Internal Reflection (FTIR) → Oldest and most widely used sensing technology.
- ▶ Uses a glass prism, light source, and CCD/CMOS camera.
- ▶ The light rays enter from the left side and exit from the right side of the prism and are focused through a lens onto a CCD or CMOS image sensor.
- ▶ Fingerprint ridges absorb light, while valleys reflect light, producing the fingerprint image.
- ▶ Reliable and widely adopted.



Fingerprint Sensors using RF imaging Techniques

b) Solid-State (Silicon) Sensors

- ▶ Introduced commercially in the mid-1990s.
- ▶ Consist of an array of tiny sensing pixels.
- ▶ No optical components are required.
- ▶ Convert fingerprint information into electrical signals
 - ▶ Capacitive sensing
 - ▶ Thermal sensing
 - ▶ Electric field sensing
 - ▶ Piezoelectric sensing
- ▶ Commonly used in smartphones and portable devices.



Fingerprint Sensors using RF imaging Techniques

c) Capacitive Sensors

- ▶ Relies on measuring the electrical capacitance created between the sensor and a user's finger.
- ▶ Made of a conductive grid of pixel cells, each acting as a tiny capacitor.
- ▶ The process:
 - Generation of an electric field
 - Detection of charge variations
 - Creation of a digital image
- ▶ Features:
 - Security: difficult to spoof
 - Design: Small and ideal for integration into mobile devices
 - Challenges: dry or wet fingers can interfere with sensor's ability and hence higher rejection rates



Fingerprint Sensors using RF imaging Techniques

d) Ultrasonic RF Sensors

- ▶ Very high frequency sound waves used to create a detailed 3-D map of a fingerprint.
- ▶ Operate using ultrasonic sound waves.
- ▶ Uses inaudible sound waves to create a highly accurate 3D map of a fingerprint.
- ▶ Sound waves penetrate the skin and create fingerprint images from reflected echoes.
- ▶ Can capture fingerprints beneath the skin surface.
- ▶ Very high frequency sound waves used to create a detailed 3-D map of a fingerprint.
- ▶ Process: Transmits sound waves, Captures the echo, Builds a 3-D image
- ▶ Has superior accuracy and works in tough conditions.
- ▶ Still developing and not widely used for large-scale deployment.
- ▶ Features: security – fake-resistant, durability and accuracy, design and integration – can operate through glass

Fingerprint Sensors using RF imaging Techniques

e) Thermal Sensors

- ▶ Works by detecting temperature differences that happen when two surfaces touch each other.
- ▶ Contains a special material called a pyroelectric material, which has a unique property → When the temperature changes, it produces a small electric current (or voltage).
- ▶ When the sensor comes in contact with something (like a finger), the temperature difference causes the material to create a voltage signal — and the sensor uses this signal to detect the contact.

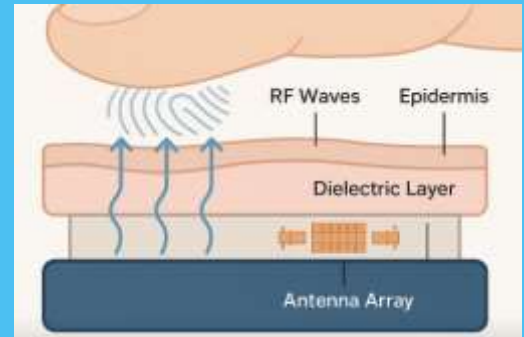
Fingerprint Sensors using RF imaging Techniques

Emerging Fingerprint Sensing Technologies and Fingerprint Quality

- ▶ Multispectral imaging captures fingerprint information from multiple skin layers.
- ▶ 3D touchless fingerprint acquisition captures fingerprints without physical contact.
- ▶ These technologies help overcome limitations such as:
 - Difficulty with wet or dry fingers.
 - Skin distortion caused by finger pressure.
 - Detection of fake or spoof fingerprints.

Fingerprint Sensors using RF imaging Techniques

- ▶ Fingerprint sensors using RF (Radio Frequency) imaging techniques rely on **electromagnetic waves** instead of just optical or capacitive methods.
- ▶ RF fingerprint sensors use radio frequency signals to penetrate the outer layer of skin (epidermis) and capture the fingerprint pattern from inner layers (dermis).
- ▶ Different from traditional optical or capacitive sensors (they only read the surface pattern).



Fingerprint Sensors using RF imaging Techniques

Components of RF Fingerprint Sensors

RF Emitter (Transmitter)

- Generates low-power radio frequency waves (typically in the range of MHz to GHz).
- Sends the RF signal toward the finger placed on the sensor surface.

RF Receiver (Antenna or Sensor Array)

- Detects the reflected RF signals that bounce back from different layers of the skin
- Measures differences in reflection caused by ridges and valleys.

Antenna Array

- A grid of micro-antennas is used for transmitting and receiving RF signals.
- Helps in capturing a detailed 2D/3D map of the subsurface structure.

Dielectric Layer

- Protective layer between the finger and the antenna.
- Designed to allow RF waves to pass through but protect electronics.

Fingerprint Sensors using RF imaging Techniques

Components of RF Fingerprint Sensors

Signal Processing Unit

- Converts the RF reflection data into a digital image.
- Uses algorithms to extract minutiae points (ridge endings, bifurcations).

Microcontroller / Processing Chip

- Handles the data analysis, encryption, and communication with the main device (phone, security system).

Power Management Unit

- Regulates power for RF transmission and signal processing.
- Important because RF imaging needs slightly higher energy than capacitive sensors.

Interface Module

- Connects the sensor with the host system (e.g., smartphone, access control).
- Often supports SPI, I²C, or proprietary protocols.

Fingerprint Sensors using RF imaging Techniques

Advantages of RF Sensors over Optical / Capacitive Sensors

- ▶ Environmental Resilience → Works with Dirty, Wet, or Damaged Fingers (since it scans deeper layers).
- ▶ Superior Anti-Spoofing → Cannot be easily fooled by surface replicas or fake fingerprints.
- ▶ Better Accuracy → Reads subsurface ridge structure

Applications of RF Sensors

- ▶ High-security devices (e.g., military, government).
- ▶ Some advanced smartphones (though rare due to cost).
- ▶ Access control in harsh environments

Fingerprint Feature Processing Techniques

- ▶ Fingerprint technology → a subset of biometric authentication → utilizes the unique patterns (unique patterns of ridges, whorls, and loops) to authenticate and verify identities in various applications.
- ▶ Enables secure access to devices, buildings, and sensitive information → by capturing and analyzing the unique characteristics of an individual's fingerprint, this technology.
- ▶ FRS employ sophisticated **algorithms** and **sensors** to scan and match fingerprints → offers a **reliable** and convenient method of biometric authentication → preferred choice for biometric authentication.
- ▶ **Benefits**
 - Ensures reliability
 - Convenient for users
 - Faster
 - Versatile
 - Ensures secure access to digital services and protecting sensitive information

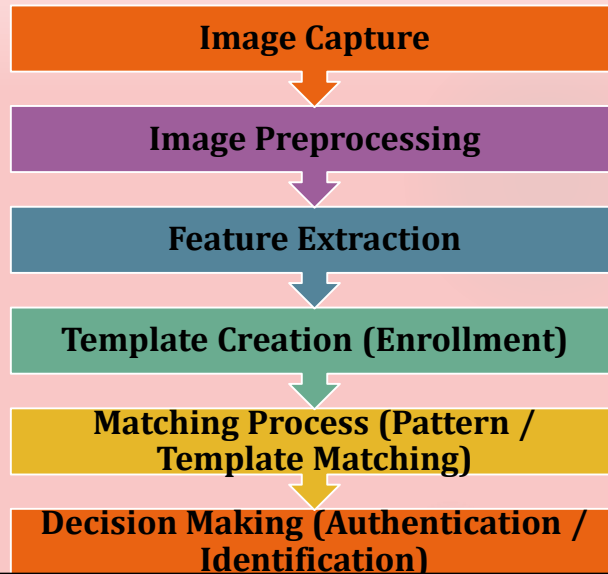
Fingerprint Feature Processing Techniques

Working of Fingerprint Technology

- ▶ Fingerprint technology relies on the unique patterns and ridges found on an individual's fingers to verify their identity.
- ▶ The process
 1. **Capturing** an image of the fingerprint → using specialized scanners
 2. **Extracting** features → Specific features such as ridge endings and bifurcations.
 3. **Creating** a digital template representing the fingerprint biometric
 4. **Forming** a mathematical model of its unique attributes.
 5. **Matching** - When a user attempts authentication,
 - a) A new fingerprint image is captured and compared to stored biometric templates using pattern-matching algorithms.
 - b) Similarities and differences between the templates analyzed using pattern matching algorithms to determine authentication.
 6. **Decision** is made → Access if a close match is found, and denied otherwise.
- ▶ This method offers high accuracy and security in various applications, including security systems and mobile devices.

Fingerprint Feature Processing Techniques

The Fingerprint Recognition Process Steps



Fingerprint Feature Processing Techniques

Processing Steps in Fingerprint Recognition

1. Image Capture:

- ▶ The first step → to capture an image of the fingerprint.
- ▶ Done using specialized fingerprint scanners, which may utilize different technologies such as optical, capacitive, or ultrasound.

2. Image Preprocessing:

- ▶ Enhances the image quality.
- ▶ Includes noise removal, image enhancement, normalization, segmentation, and thinning.
- ▶ This step is present in most practical fingerprint recognition systems.

3. Feature Extraction:

- ▶ Once the fingerprint image is captured, the system extracts specific features from it.
- ▶ Features include ridge endings, bifurcations, core point, delta point, ridge orientations and other unique characteristics of the fingerprint.

Fingerprint Feature Processing Techniques

Processing Steps in Fingerprint Recognition

4. Template Creation:

- ▶ The extracted features are then used to create a digital template or representation of the fingerprint.
- ▶ The template → a mathematical model that captures the unique attributes of the fingerprint

5. Matching Process:

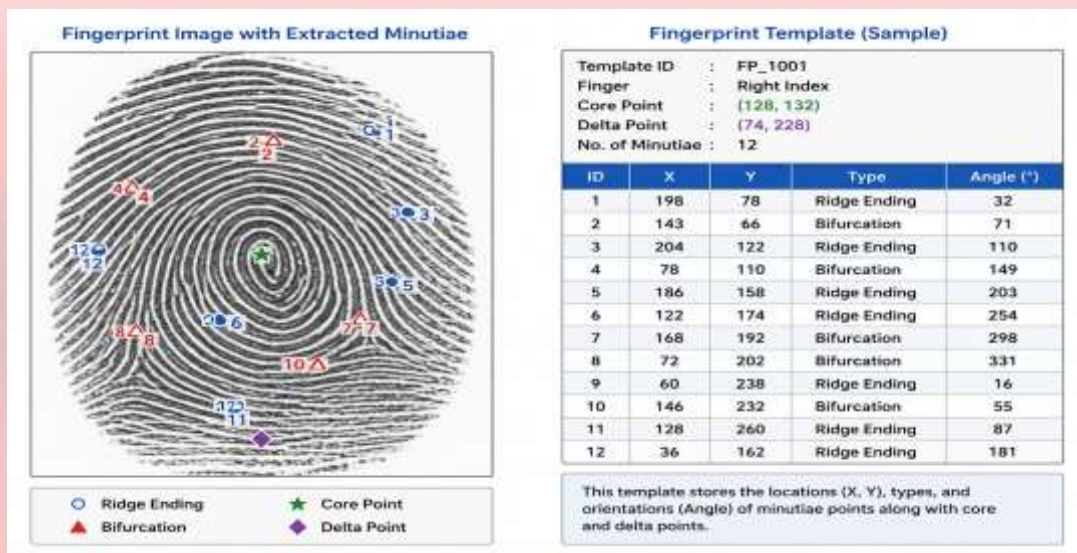
- ▶ When a user attempts to authenticate their identity using their fingerprint, the system captures a new image of the fingerprint → probe image / query.
- ▶ This probe image is processed to create a new template, which is then **compared** to the stored templates in the system's database.
- ▶ The system employs pattern-matching algorithms to compare the new template with the stored templates.
- ▶ Looks for similarities and differences between the two templates to determine whether they match. Calculates a **similarity score**.

6. Decision Making:

- ▶ Based on the results of the pattern-matching process, the system decides the authenticity of the fingerprint.
- ▶ If the new template closely matches any of the stored templates, the user is granted access. Otherwise, access is denied.

Fingerprint Feature Processing Techniques

Fingerprint Template



Fingerprint Feature Processing Techniques

Fingerprint Acquisition (Capture)

- ▶ **Image acquisition** - the process of capturing and digitizing the fingerprint of an individual for further processing.
- ▶ Ink-on-paper technique → used to capture the fingerprint information from a person → a simple - yet powerful - acquisition and recording technique → used since the late 19th century.
- ▶ Reason for the popularity of fingerprint recognition → availability of mature, convenient, and low-cost sensors that can rapidly acquire the fingerprint of an individual with minimum or no intervention from a human operator.
- ▶ Used in laptops and mobile phones.
- ▶ Fingerprint Sensing Techniques → 1. off-line technique 2. on-line technique

Fingerprint Feature Processing Techniques

Fingerprint Sensing Techniques – 1. Offline Techniques

- ▶ Do not produce the digital image directly from the fingertip.
- ▶ 2 offline techniques
- ▶ A) Ink-paper method
 - Fingerprint is first transferred to a substrate that is subsequently digitized.
 - By applying ink to the subject's fingertip and then rolling or pressing the finger on paper → an impression of the fingerprint ridges created on paper.
 - The impression is then scanned and digitized using a flatbed document scanner.
 - Infeasible and socially unacceptable in the context of non-forensic applications
- ▶ B) Latent fingerprint method
 - Lifted from the surface of objects that are touched or handled by a person – using photography / dusting / chemical processing

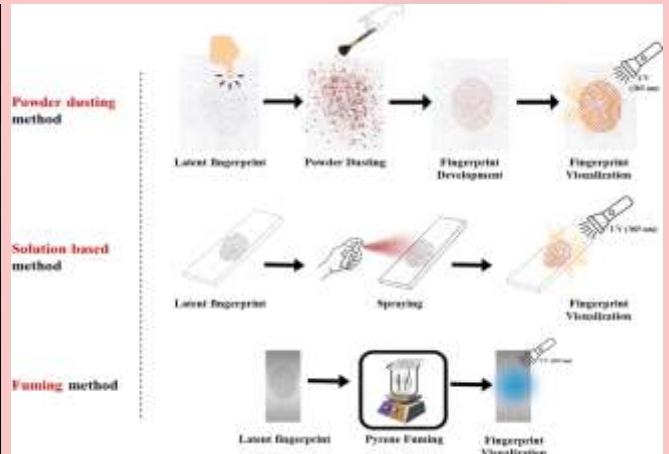
Fingerprint Feature Processing Techniques

Fingerprint Sensing Techniques – 1. Offline Techniques

A) Ink-paper method



B) Latent fingerprint method



Fingerprint Feature Processing Techniques

Fingerprint Sensing Techniques – 2. Online Techniques

- ▶ Digital image produced directly from a subject's fingertip via digital imaging technology → live-scan fingerprint.
- ▶ Few online techniques
 - 1. Optical Frustrated Total Internal Reflection
 - 2. Capacitance based sensing
 - 3. Ultrasound based sensing
 - 4. Temperature based sensing
 - 5. RF based sensing

Fingerprint Feature Processing Techniques

Image Quality

- ▶ Precise extraction of fingerprint features is crucially important.
- ▶ A good quality fingerprint image → has **distinguishable patterns and features** that allow the extraction of features that are useful for subsequent matching of fingerprint pairs.
- ▶ Factors that influence fingerprint image quality → environmental, behavioral, quality of biometrics presented, system ability, wet fingers, operating the equipment in extreme environmental conditions (Environmental factor), scratched or smudged scanning surface.

Fingerprint Feature Processing Techniques

Efforts for improving the quality of the image

- ▶ Live scan or electronic capture of fingerprint
 - Faster and efficient - produces clearer, sharper and high quality fingerprint images over ink-and-roll method.
- ▶ Controlling environmental conditions, user behavior and familiarizing users with acquisition process
 - For example, rubbing dirty fingertips with alcohol or drying up wet fingers will result in improved quality of sampled data.
 - Clean and scratch free scanning surface.
 - Avoiding unnecessary disturbances or extreme environmental conditions.
- ▶ Image quality checks
 - To reject poorly scanned images.
 - Using threshold based recognition algorithm → to accept or reject the sampled data and ask for re-sampling.

Fingerprint Feature Processing Techniques

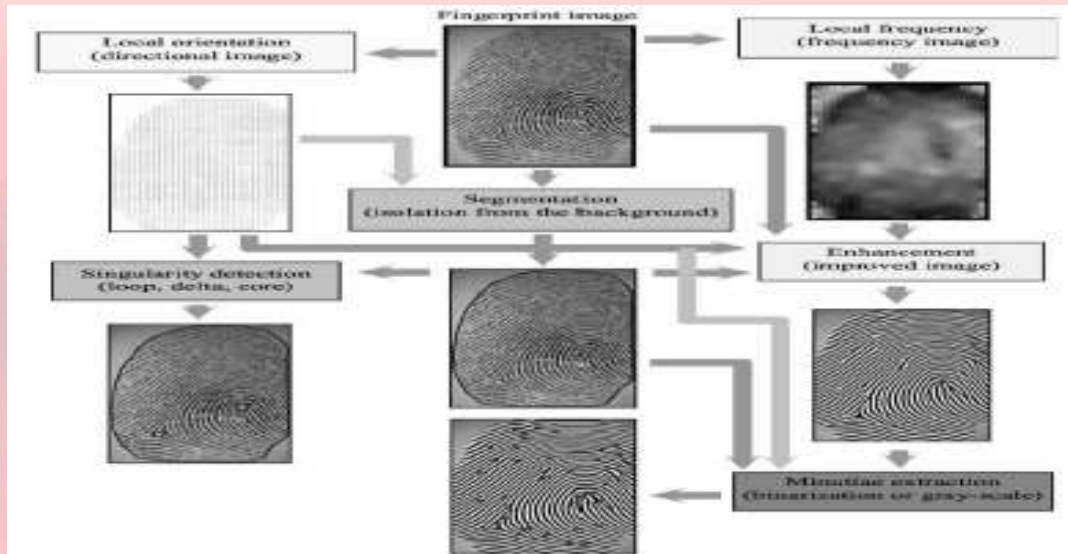
- ▶ Fingerprint feature processing techniques → the methods used to **extract**, **represent**, and **match** unique fingerprint characteristics for biometric identification and verification.
- ▶ Improves recognition accuracy while reducing computational complexity.
- ▶ Few techniques:
- ▶ **Image Enhancement**
 - The captured fingerprint image is enhanced to improve ridge clarity and remove noise
 - Purpose → to improve ridge-valley contrast, to remove noise and distortions, to increase feature extraction accuracy
 - Methods → Histogram Equalization, Gabor Filtering, Fourier Transform Enhancement, Median Filtering, Wiener Filtering
- ▶ **Image Binarization**
 - Converts a grayscale fingerprint image into a binary image → Ridge pixels → Black (1) and Valley pixels → White (0)
 - Methods → Global Thresholding, Adaptive Thresholding, Otsu's Method
 - Advantages → Simplifies feature extraction, reduces storage requirements

Fingerprint Feature Processing Techniques

- ▶ **Image Thinning (Skeletonization)**
 - Reduces ridge thickness to a single-pixel width while preserving ridge structure.
 - Purpose → For easy minutiae extraction and accurate ridge analysis
 - Methods → Zhang-Suen Algorithm, Guo-Hall Algorithm, Morphological Thinning
- ▶ **Minutiae Feature Extraction**
 - Extracting minutiae features like ridge bifurcation, short ridge, bridge, island, dot, spur, etc.
 - Used in criminal identification, attendance systems, mobile authentication, etc.
 - Advantages → Simplifies feature extraction, Reduces storage requirements.
- ▶ **Ridge Feature Extraction**
 - Instead of individual minutiae, ridge characteristics are extracted → ridge orientation, ridge frequency, ridge count, ridge density, ridge shape.
 - Applied for low-quality fingerprint matching and partial fingerprint recognition

Fingerprint Feature Processing Techniques

► Fingerprint Feature Extraction Steps



Minutiae Extraction

- Fingerprint feature extraction techniques → the methods used to identify unique characteristics from a fingerprint image for recognition.
- Classified into the following categories:

1. Minutiae-Based Techniques

- Most widely used - Minutiae are small details in fingerprint ridges, such as ridge endings, bifurcations, etc.

2. Pattern-Based (Global) Techniques

- Uses the overall structure of the fingerprint rather than fine details.

3. Image-Based (Texture Analysis) Techniques

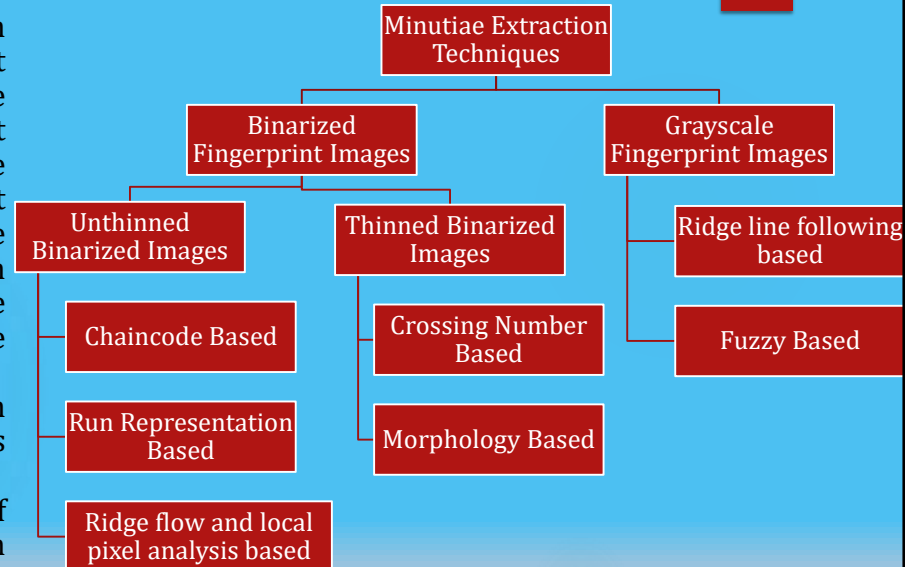
- Extracts features directly from the gray-level image without binarization.

4. Hybrid Techniques

- Combination of minutiae and texture features to improve robustness.

Minutiae Extraction

- ▶ Minutiae extraction technique → most widely used technique of fingerprint representation → the backbone of most currently available fingerprint recognition products → more accurate → the template size is smaller.
- ▶ Two fingerprints match if their minutiae points match.
- ▶ Classification of minutiae extraction techniques:



Minutiae Extraction

- ▶ **Minutiae extraction** → the process of identifying unique fingerprint features such as ridge endings and ridge bifurcations, which are used for fingerprint matching and identification.
- ▶ Two categories:
 1. Minutiae extraction from binarized fingerprint images
 2. Minutiae extraction from grayscale fingerprint images

Minutiae Extraction

1. Binarized fingerprint images

- ▶ Binarized fingerprint images → the result of converting a grayscale fingerprint image into a binary format → each pixel is either black (ridge) or white (valley) → Before minutiae extraction, the grayscale fingerprint is converted into a binary image.
- ▶ Classified as:
 - A. Unthinned binarized images
 - B. Thinned binarized images

Minutiae Extraction

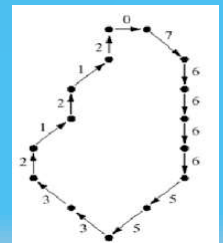
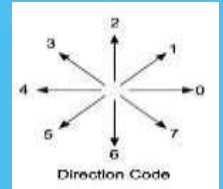
1. A. Unthinned Binarized Images

- ▶ Ridges are still several pixels wide → The ridge thickness has not been reduced.
- ▶ Advantages
 - Less preprocessing
 - Preserves ridge information
 - Faster preprocessing
- ▶ Disadvantages
 - More difficult to locate exact ridge endings
 - Higher number of false minutiae
- ▶ 3 methods → i) Chaincode based technique,
ii) Run representation-based technique,
iii) Ridge flow and local pixel analysis-based technique.

Minutiae Extraction

1. A. Unthinned Binarized Images - i) Chaincode Based Technique

- ▶ Chain code represents the boundary of a ridge (or ridge contours) as an array of directional vectors.
- ▶ Each movement is encoded using numbers (0-7).
- ▶ The system scans the binarized image and traces the boundary of each black fingerprint ridge.
- ▶ Each step along the contour is encoded into a sequence of numbers (such as Freeman 0-7 directions) showing the path of the edge.
- ▶ Sharp shifts, turning points, or sudden breaks in the chaincode sequence point directly to a minutia (ending or split) → Feature detection.
- ▶ Gives good boundary representation and accurate ridge tracking
- ▶ But Sensitive to noise and computationally expensive



Minutiae Extraction

1. A. Unthinned Binarized Images - ii) Run Representation-based Technique

- ▶ Instead of examining every pixel, continuous ridge pixels are grouped into runs.
- ▶ Eg. 000111110000111000 → Run1 = pixels 4-8, Run2 = pixels 13-15
- ▶ Extracts fingerprint minutiae directly from binary images using run-length encoding (RLE), skipping the slow image-thinning step.
- ▶ Converts the raw grayscale fingerprint image into a clean black-and-white binary image.
- ▶ Scans the binary image row-by-row or column-by-column to group continuous black pixels into discrete segments called "runs" → Run Length Encoding (RLE)
- ▶ Analyzes the adjacency relationships between runs in neighboring rows or columns to find structural anomalies → starting run, ending run, merging run, splitting run.
- ▶ Applies predefined geometric constraints and spatial checks to filter out noise, spikes, and false triggers.
- ▶ Requires less memory and faster processing but less accurate.

Minutiae Extraction

1. A. Unthinned Binarized Images - iii) Ridge Flow And Local Pixel Analysis-based Technique

- ▶ This method combines ridge orientation and local neighborhood analysis.
- ▶ The algorithm studies ridge direction, ridge continuity and pixel arrangement.
- ▶ The ridge orientation is estimated.
- ▶ The local neighborhood is examined.
- ▶ Abrupt changes are identified.
- ▶ Minutiae are marked.
- ▶ Offers better accuracy but requires more computations and sensitive to orientation estimation errors

Minutiae Extraction

1. B. Thinned Binarized Images

- ▶ Before minutiae extraction, the fingerprint undergoes thinning (skeletonization).
- ▶ Each ridge becomes only one-pixel wide.
- ▶ Advantages
 - Easy minutiae detection
 - High accuracy
- ▶ Disadvantages
 - Thinning may introduce artifacts
 - False minutiae if thinning is poor
- ▶ 2 methods → i) Crossing Number based technique
ii) Morphology based technique

Minutiae Extraction

1. B. Thinned Binarized Images - i) Crossing Number Based Technique

- ▶ Most widely used minutiae extraction technique.
- ▶ A 3×3 neighborhood around every ridge pixel is examined.
- ▶ Crossing number, $CN = \frac{1}{2} \sum_{i=1}^8 |P_i - P_{i+1}|$ where $P_9 = P_1$.
- ▶ The CN indicates the type of ridge point.
- ▶ Crossing Number Feature
- ▶ 0 → Isolated point, 1 → Ridge ending, 2 → Normal ridge point, 3 → Ridge bifurcation and 4 → Crossing
- ▶ Eg. Only one right neighbor for P → P1
- ▶ 3 neighbors for P → P1, P3, P5

P3	P2	P1
P4	P	P8
P5	P6	P7

$$CN = \frac{1}{2} \sum_{i=1}^8 |P_i - P_{i+1}|$$

Minutiae Extraction

Eg. 1: Only the right neighbor (P1) is 1, others 0.

Neighbors P1..P8 = [1,0,0,0,0,0,0,0]

Append P1 at end to form circular sequence: [1,0,0,0,0,0,0,0,1]

Compute absolute differences:

$$|P1-P2| = |1-0| = 1$$

$$|P2-P3| = |0-0| = 0$$

$$|P3-P4| = |0-0| = 0$$

$$|P4-P5| = |0-0| = 0$$

$$|P5-P6| = |0-0| = 0$$

$$|P6-P7| = |0-0| = 0$$

$$|P7-P8| = |0-0| = 0$$

$$|P8-P9| = |0-1| = 1$$

$$\text{Sum} = 1 + 0 + 0 + 0 + 0 + 0 + 0 + 1 = 2$$

$$CN = 0.5 \times 2 = 1 \rightarrow \text{Ridge Ending}$$

P3	P2	P1
P4	P	P8
P5	P6	P7

Eg. 2: 3 neighbors (P1, P3, P5) are 1, others 0.

Neighbors P1..P8 = [1,0,1,0,1,0,0,0]

Append P1 at end to form circular sequence: [1,0,1,0,1,0,0,0,1]

Compute absolute differences:

$$|P1-P2| = |1-0| = 1$$

$$|P2-P3| = |0-1| = 1$$

$$|P3-P4| = |1-0| = 1$$

$$|P4-P5| = |0-1| = 1$$

$$|P5-P6| = |1-0| = 1$$

$$|P6-P7| = |0-0| = 0$$

$$|P7-P8| = |0-0| = 0$$

$$|P8-P9| = |0-1| = 1$$

$$\text{Sum} = 1 + 1 + 1 + 1 + 1 + 0 + 0 + 1 = 6$$

$$CN = 0.5 \times 6 = 3 \rightarrow \text{Ridge bifurcation}$$

P3	P2	P1
P4	P	P8
P5	P6	P7

Minutiae Extraction

1. B. Thinned Binarized Images - ii) Morphology Based Technique

- ▶ Uses mathematical morphology (or) mathematical shapes called structuring elements to process image ridges and valleys.
- ▶ Morphological operations include dilation, erosion, opening and closing
- ▶ These operations remove spikes, bridges, holes and noise.
- ▶ The fingerprint is thinned
- ▶ Morphology is applied.
- ▶ Ridge endings are detected.
- ▶ Bifurcations are detected.
- ▶ Faster processing with improved accuracy
- ▶ The process depends on structuring element

Minutiae Extraction

2. Grayscale fingerprint images

- ▶ An image where each pixel represents a shade of gray (0 to 255), instead of color.
- ▶ The most common format for fingerprint processing before binarization.
- ▶ The methods work directly on the original grayscale fingerprint without converting it into a binary image.
- ▶ Preserves original ridge information
- ▶ More accurate technique
- ▶ Less affected by binarization errors
- ▶ But has higher computational complexity
- ▶ 2 methods:
 - A. Ridge Line Following-based Technique
 - B. Fuzzy-based Technique

Minutiae Extraction

2. A. Ridge Line Following-based Technique

- ▶ The algorithm traces ridge centers directly in the grayscale image.
- ▶ Instead of thinning, it follows the ridge using ridge orientation, ridge intensity and ridge continuity
- ▶ Enhances the image.
- ▶ Estimates ridge orientation.
- ▶ Detects ridge center.
- ▶ Follows ridge.
- ▶ Locate endings and bifurcations.
- ▶ Accurate and false minutiae reduced.
- ▶ Computationally intensive

Minutiae Extraction

2. B. Fuzzy-based Technique

- ▶ Fingerprint images often contain uncertainty due to noise, smudges, or poor-quality impressions. Fuzzy logic **handles this uncertainty** by assigning each pixel a degree of membership (between 0 and 1) instead of classifying it as simply ridge or valley.
- ▶ Enhance the grayscale fingerprint.
- ▶ Compute local features such as intensity, ridge orientation, and continuity.
- ▶ Apply fuzzy membership functions to determine the **likelihood** that a pixel belongs to a ridge.
- ▶ Use **fuzzy rules** to identify ridge endings and bifurcations.
- ▶ Remove false minutiae using fuzzy decision-making.
- ▶ Handles noisy and low-quality fingerprints well.
- ▶ Designing effective fuzzy rules and membership functions can be challenging.

Minutiae Extraction

2. B. Fuzzy-based Technique

- ▶ The finger may have oil, dust, and sweat.
- ▶ Also the captured fingerprint contains broken ridges, blurred regions, and noise.
- ▶ A Fuzzy-based method assigns confidence values such as below to identify nature of the ridge:

Pixel/Region	Membership to Ridge
Clear ridge	0.98
Slightly blurred ridge	0.82
Smudged ridge	0.65
Very noisy area	0.35
Valley	0.05

Minutiae Extraction

2. B. Fuzzy-based Technique – The process

- A conventional algorithm might incorrectly identify several false ridge endings when the ridges appear broken.
- A fuzzy-based algorithm recognizes that the broken parts still have a **high probability** of belonging to the same ridge, reconnects them logically, and extracts the correct minutiae

Captured Fingerprint

Noisy / Smudged Fingerprint

Compute Membership Values (Ridge = 0 to 1)

Apply Fuzzy Rules

Identify Reliable Ridges

Computer Enhancement and Modelling of Fingerprint Images

Fingerprint Image Enhancement

- ▶ Improves the quality of a captured fingerprint so that ridges and valleys can be clearly identified and features such as minutiae can be extracted accurately.
- ▶ Common Enhancement Techniques
 - **Normalization:** Adjusts the brightness and contrast of the fingerprint image.
 - **Noise Removal:** Removes unwanted noise and artifacts.
 - **Segmentation:** Separates the fingerprint region from the background.
 - **Ridge Enhancement:** Makes the ridge and valley patterns clearer.
 - **Filtering:** Uses filters such as Gabor filters to enhance ridge structures.
 - **Binarization:** Converts the grayscale image into a binary image containing ridges and valleys.
 - **Thinning:** Reduces fingerprint ridges to approximately one-pixel width for easier minutiae extraction.

Computer Enhancement and Modelling of Fingerprint Images

Fingerprint Image Modelling

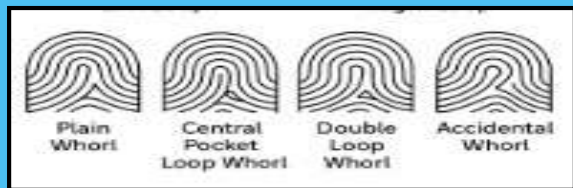
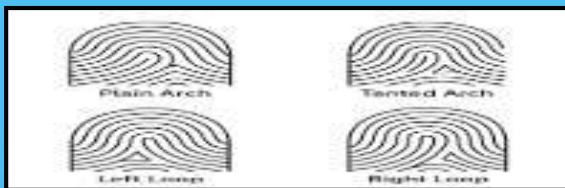
- ▶ Represents the fingerprint using mathematical or computational models that describe its important characteristics..
- ▶ The model may represent:
 - ▶ Ridge orientation
 - ▶ Ridge frequency
 - ▶ Ridge density
 - ▶ Minutiae positions
 - ▶ Local texture
 - ▶ Core and delta points
- ▶ Improves fingerprint image quality.
- ▶ Helps extract reliable features.
- ▶ Reduces the effect of noise and poor finger placement.
- ▶ Improves matching accuracy and recognition performance.

Fingerprint Classification

- ▶ **Classification** → stage that comes after feature extraction and before matching/identification.
- ▶ **Definition** → Fingerprint classification is the process of categorizing fingerprints into predefined pattern classes based on the overall ridge flow.
- ▶ Helps reduce the search space during fingerprint matching by comparing a fingerprint only with those in the same class.
- ▶ **Role** → to categorize the fingerprint into one of the standard fingerprint classes so that the search space is reduced during identification. → Assign the fingerprint into a **class based on ridge flow pattern**.
- ▶ **Need for fingerprint classification**
 - ▶ Speeds up fingerprint identification.
 - ▶ Reduces the search time in large fingerprint databases.
 - ▶ Improves the efficiency of Automated Fingerprint Identification Systems (AFIS).
 - ▶ Organizes fingerprint records systematically.

Fingerprint Classification

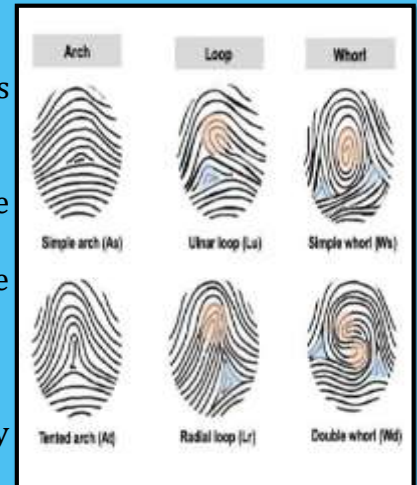
- ▶ 8 sub-classes of fingerprint



- ▶ The four different whorl classes can be combined as **Whorl (W) class**.
- ▶ The PA and TA classes can be combined as the **Arch (A) class**
- ▶ The Left Loop and Right Loop can be considered as **L and R classes**
- ▶ The natural frequencies of classes: W – 27.9%, L – 33.8%, R – 31.7% and A – 6.6%

Fingerprint Classification

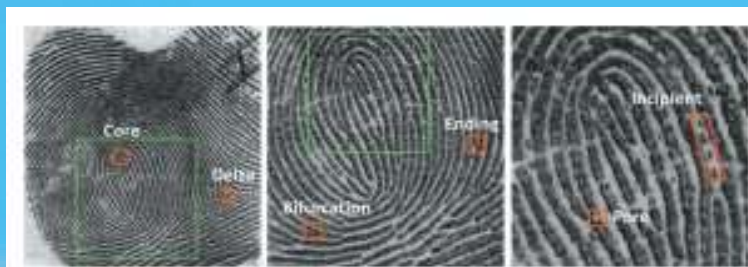
- ▶ Classification done based on the differences →
 - ▶ in terms of the global flow patterns of the ridge curves.
 - ▶ in terms of the number and locations of singular points in the fingerprint image
 - ▶ Examples:
 - ▶ LL – exactly one core and one delta; the core is to the left of the delta
 - ▶ RL – exactly one core and one delta; the core is to the right of the delta
 - ▶ W – Two cores and two deltas
 - ▶ PA – no singular points
 - ▶ TA – one core and one delta; the delta is approximately directly below the core.
- ▶ 2 approaches → Traditional (rule-based) approach, Modern (machine learning/deep learning) approach



Fingerprint Classification - Approaches

1. Traditional (rule-based) approach

- ▶ Classification is done using explicit rules derived from fingerprint features such as singular points, ridge orientation, and ridge counts.
- ▶ Uses ridge flow patterns, singular points (core and delta), and geometric characteristics.
- ▶ Eg. Point Care Index method – to detect singular points(Core and Delta)



Fingerprint Classification - Approaches

1. Traditional (rule-based) approach

- ▶ Significance of singular points (core and delta) → critical for
 - ▶ Fingerprint classification (arch, loop, whorl, etc.)
 - ▶ Alignment of fingerprints before minutiae matching
 - ▶ Reliable feature extraction
- ▶ **Point Care Index Method** → a mathematical approach used to locate these singular points by analyzing the orientation field of the fingerprint ridges.
- ▶ **Steps:**
 1. Computation of orientation field using gradient method
 2. Computation of point care index using orientation differences
 3. Detection (or Localization) of singular points using the PCI
 4. Classification of the fingerprint

Fingerprint Classification - Approaches

Point Care Index (PCI) Method

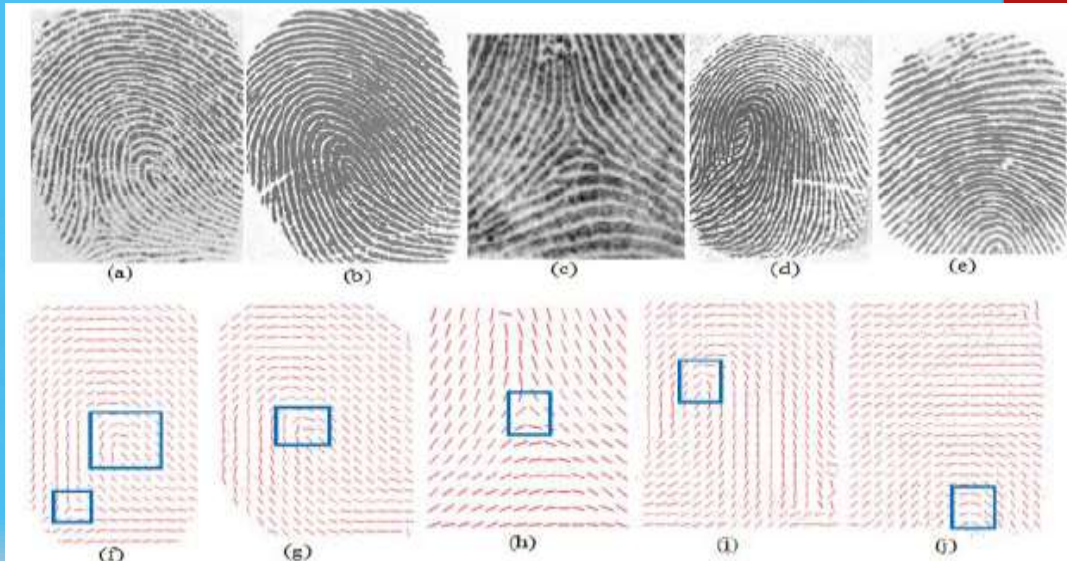
Step-1: Computation of orientation field using gradient method

- ▶ A fingerprint image is divided into non-overlapping blocks (commonly 8×8 or 16×16 pixels).
- ▶ For each block, the ridge orientation $\theta(i,j)$ is computed using gradients (an iterative mathematical technique used to find the minimum or maximum of a function):

$$\theta(i,j) = \frac{1}{2} \tan^{-1} \left(\frac{2 \sum G_x G_y}{\sum (G_x^2 - G_y^2)} \right)$$

- ▶ Here G_x and G_y → image gradients along x and y

Fingerprint Classification - Approaches



Fingerprint Classification - Approaches

Step-2: Computation of point care index using orientation differences

- The Pointcare Index (PI) → measures how much the orientation changes around a closed loop (usually 8 blocks around a central block).

$$PI = \frac{1}{2\pi} \sum_{k=1}^n \Delta\theta_k$$

- $n = 8$ (neighbors in a loop)
- $\Delta\theta_k = \theta_{k+1} - \theta_k$ (difference in orientation between adjacent neighbors)
- Adjust $\Delta\theta_k$ to be within $(-\pi/2, +\pi/2)$ to avoid jumps.

- Interpretation of values:

PI	PATTERN	FREQUENCY
+0.5 or +180°	Core point	• Found in loop and whorl patterns. • Represents a ridge "turning" center.
-0.5 or -180°	Delta point	• Found in loop and whorl patterns. • Represents a ridge diverging point
0	No singularity (plain region)	
1	Double core or rare singularities	

Fingerprint Classification - Approaches

Step-3: Detection (or Localization) of singular points using the PCI

- Consider 3 x 3 block

$$\theta_1 = 10^\circ, \theta_2 = 40^\circ, \theta_3 = 80^\circ, \theta_4 = 120^\circ, \theta_5 = 160^\circ, \theta_6 = -160^\circ, \theta_7 = -120^\circ, \theta_8 = -80^\circ$$

θ_1	θ_2	θ_3
θ_8	C	θ_4
θ_7	θ_6	θ_5

- To calculate PI

$$PI = \frac{1}{2\pi} \sum_{k=1}^n \Delta\theta_k$$

We compute $\Delta\theta = \theta_{k+1} - \theta_k$, making sure each difference stays within $(-90^\circ, +90^\circ)$.

- $\Delta\theta_1 = 40 - 10 = +30^\circ$
- $\Delta\theta_2 = 80 - 40 = +40^\circ$
- $\Delta\theta_3 = 120 - 80 = +40^\circ$
- $\Delta\theta_4 = 160 - 120 = +40^\circ$
- $\Delta\theta_5 = (-160) - 160 = -320 \rightarrow \text{adjust} \rightarrow +40^\circ$
- $\Delta\theta_6 = (-120) - (-160) = +40^\circ$
- $\Delta\theta_7 = (-80) - (-120) = +40^\circ$
- $\Delta\theta_8 = 10 - (-80) = +90^\circ$

$$\text{Total change} = 30 + 40 + 40 + 40 + 40 + 40 + 40 + 90 = 360^\circ$$

$$PI = \frac{1}{2\pi} \sum \Delta\theta = \frac{360^\circ}{360^\circ} = +1$$

Core point is detected at the centre

Fingerprint Classification - Approaches

2. Modern (machine learning/deep learning based) approach

- Statistical Methods
- Machine learning approaches

Statistical Methods

- The fingerprint image is represented as numerical data (feature vectors)
- Eg. A list of points [x, y, angle, type] → Point 1: [150, 220, 45 degrees, Ridge Ending]
- Statistical decision-making is applied to classify the fingerprint into a class (arch, loop, whorl, etc.).
- Instead of hard rules, they use mathematical models, probability, and similarity measures.
- Eg. Ridge orientation analysis, Ridge frequency analysis and ridge count.

Fingerprint Classification - Approaches

Machine Learning approaches

- ▶ Unlike rule-based (explicit rules) or statistical (feature distributions), machine learning (ML) methods learn classification rules from data.
- ▶ Features (like orientation, ridge frequency, singular points, or even raw images) are considered and a model is trained to predict the class (arch, loop, whorl, etc.).
- ▶ Eg. algorithms → K-nearest neighbor (KNN), support vector machine (SVM), Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN)

Fingerprint Classification - Approaches

Machine Learning approaches – KNN Algorithm

- ▶ KNN → a non-parametric supervised learning technique → used to classify the data point to a given category with the help of training set → and classifies new cases based on a similarity.
- ▶ Also called an **instance-based learning algorithm** → because predictions are made based on stored instances.
- ▶ Widely used due to its simplicity and effectiveness in various applications.
- ▶ Predictions are made for a new instance (x) by searching through the entire training set for the K most similar cases (neighbors) and summarizing the output variable for those K cases.
- ▶ In classification this is the **mode** (or most common) class value.

Fingerprint Classification - Approaches

Machine Learning approaches – KNN Algorithm

► Step 1: Choose the Number of Neighbors (k)

- Select the value of k, which represents the number of nearest neighbors (fingerprint samples) are considered for classification.
- A small k may be sensitive to noise, while a large k may reduce classification accuracy.

► Step 2: Prepare the Data

- Split the dataset into training and test sets.
- Normalize or standardize the data if necessary to ensure that all features contribute equally to the distance calculations.

► Step 3: Calculate Distances

- For each test fingerprint, calculate its distance from every fingerprint in the training set using distance measures such as:

1. Euclidean distance

2. Manhattan distance

3. Minkowski distance

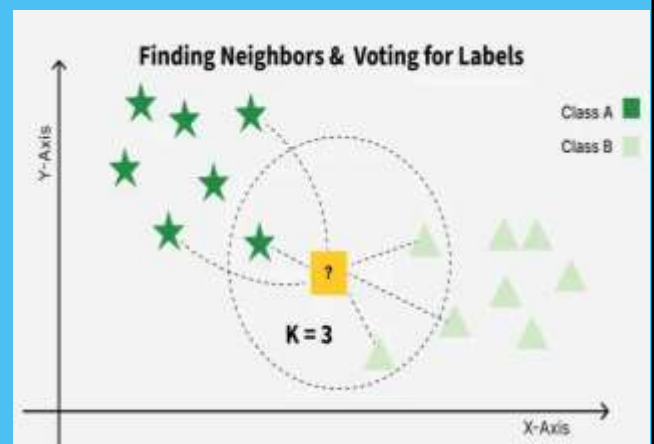
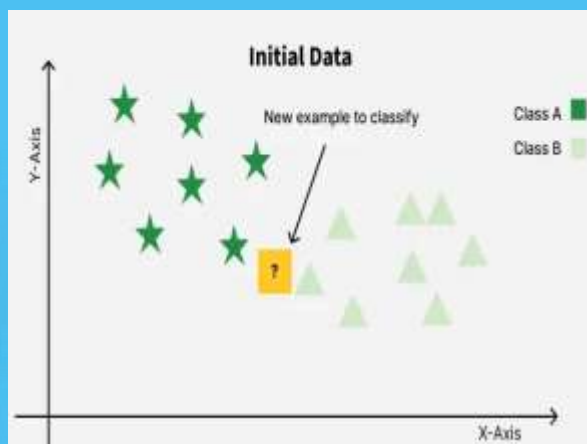
$$\text{distance}(x, X_i) = \sqrt{\sum_{j=1}^d (x_j - X_{ij})^2}$$

$$d = |x_2 - x_1| + |y_2 - y_1|$$

$$d = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$

Fingerprint Classification - Approaches

Machine Learning approaches – KNN Algorithm



Fingerprint Classification - Approaches

Machine Learning approaches – KNN Algorithm

▶ Step 4: Identify Nearest Neighbors

- ▶ Sort the calculated distances.
- ▶ Select the k fingerprints with the smallest distances.

▶ Step 5: Make Predictions / Classify the Fingerprint

- ▶ For classification:
 - ▶ Assign the class label (A / W / L / R) that is most common among the k nearest neighbors → majority voting.
- ▶ For regression:
 - ▶ Calculate the average (or weighted average) of the target values of the k nearest neighbors.

▶ Step 6: Evaluate the Model

- ▶ Assess the model's performance using metrics such as accuracy, precision, recall, F1 score, confusion matrix or mean squared error, depending on whether the task is classification or regression.

Fingerprint Classification - Approaches

Machine Learning approaches – KNN Algorithm - Example

- ▶ Consider the training dataset with 2 features: Feature 1: Number of Ridge Endings, Feature 2: Number of Bifurcations
- ▶ A new fingerprint NF has: Ridge Endings = 16 and Bifurcations = 12
- ▶ Using the KNN algorithm with k = 3 and Euclidean distance, determine the class of the new fingerprint.

Fingerprint	Ridge Endings	Bifurcations	Class
F1	12	8	Arch
F2	15	10	Loop
F3	14	11	Loop
F4	20	18	Whorl
F5	18	17	Whorl

Fingerprint Classification - Approaches

Machine Learning approaches – KNN Algorithm - Example

► Solution

- Calculate Euclidean Distance for each fingerprint F1, ..., F5 to the NF

$$\text{distance}(x, X_i) = \sqrt{\sum_{j=1}^d (x_j - X_{ij})^2}$$

Distance from F1 (Arch)

$$\begin{aligned} &= \sqrt{(16 - 12)^2 + (12 - 8)^2} \\ &= \sqrt{16 + 16} = \sqrt{32} = 5.66 \end{aligned}$$

Distance from F2 (Loop)

$$\begin{aligned} &= \sqrt{(16 - 15)^2 + (12 - 10)^2} \\ &= \sqrt{1 + 4} = \sqrt{5} = 2.24 \end{aligned}$$

Distance from F3 (Loop)

$$\begin{aligned} &= \sqrt{(16 - 14)^2 + (12 - 11)^2} \\ &= \sqrt{4 + 1} = \sqrt{5} = 2.24 \end{aligned}$$

Distance from F4 (Whorl)

$$\begin{aligned} &= \sqrt{(16 - 20)^2 + (12 - 18)^2} \\ &= \sqrt{16 + 36} = \sqrt{52} = 7.21 \end{aligned}$$

Distance from F5 (Whorl)

$$\begin{aligned} &= \sqrt{(16 - 18)^2 + (12 - 17)^2} \\ &= \sqrt{4 + 25} = \sqrt{29} = 5.39 \end{aligned}$$

Fingerprint Classification - Approaches

Machine Learning approaches – KNN Algorithm - Example

► Solution (contd...)

- Sort and identify nearest k neighbors

Fingerprint	Ridge Endings	Bifurcations	Class	Distance
F2	15	10	Loop	2.24
F3	14	11	Loop	2.24
F5	18	17	Whorl	5.39
F1	12	8	Arch	5.66
F4	20	18	Whorl	7.21

- Classify the Fingerprint based on majority voting technique → Loop – 2, Whorl – 1
- Since Loop has the majority of votes, the unknown fingerprint is classified as a **Loop**.

Fingerprint Matching

- ▶ Fingerprint matching → the process of comparing the extracted features (fingerprint template) of an input fingerprint with one or more stored fingerprint templates to determine whether they belong to the same person.
- ▶ Easier → Matching high quality fingerprints with small intra-subject variations → algorithm gives high accurate results.
- ▶ Challenge → matching samples of poor quality affected by:
 - i. Large displacement and/or rotation
 - ii. Non-linear distortion
 - iii. Different pressure and skin condition
 - iv. Feature extraction errors.
- ▶ Fingerprint matching approaches :
 1. Correlation-based matching
 2. Minutiae-based matching
 3. Ridge feature-based matching.

Fingerprint Matching

- ▶ Two different impressions of the same finger – with large intra-subject variations.
- ▶ Fingerprint images from different fingers – with small inter-subject variations (look similar).



Fingerprint Matching

1. Correlation-based Matching Technique

- ▶ Let T = Template fingerprint image (stored in the database)
- ▶ I = Input fingerprint image (captured by the scanner)
- ▶ $I(\Delta x, \Delta y, \theta)$ = Input image after shifting by Δx pixels in the x direction, shifting by Δy pixels in the y direction and rotating by an angle θ
- ▶ The similarity between the two fingerprint images is calculated using cross-correlation:

$$S(T, I) = \max_{\Delta x, \Delta y, \theta} CC(T, I(\Delta x, \Delta y, \theta))$$

- ▶ where $CC(T, I) = T^T I \rightarrow$ cross-correlation measure.

Fingerprint Matching

1. Correlation-based Matching Technique

▶ Working Principle

1. Capture the input fingerprint.
2. Rotate and shift the input fingerprint by different amounts.
3. Compute the cross-correlation between the template and transformed input image.
4. Find the maximum correlation value.
5. The transformation giving the highest correlation provides the best alignment (registration).
6. If the maximum correlation exceeds a threshold, the fingerprints are considered a match.

▶ Advantages

- Simple concept.
- Uses the entire fingerprint image.
- No need to explicitly extract minutiae points.

Fingerprint Matching

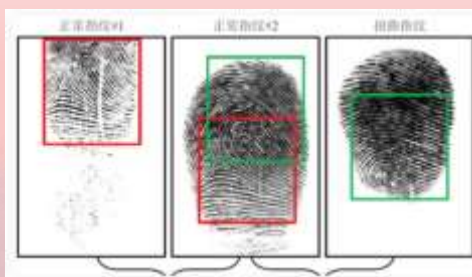
1. Correlation-based Matching Technique

► Limitations

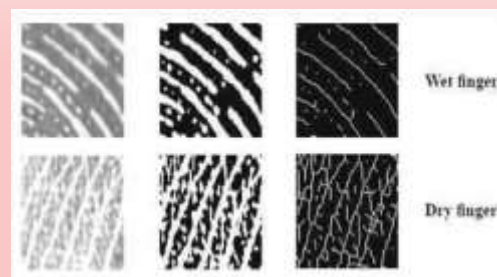
Issue	Problem	Impact	Solution
Non-linear Distortion	The same finger may produce different impressions – due to Skin elasticity, finger pressure, finger placement	local deformation, reduced effectiveness of global correlation	Use local or block-wise correlation techniques
Variations in Image Quality	Different impressions may have different brightness, contrast and ridge thickness	Reduced correlation values	Use advanced correlation measures and image enhancement techniques.
High Computational Cost	The system must test many combinations of: Rotation (θ), Horizontal shift (Δx), Vertical shift (Δy)	Requires a large number of calculations.	Use local correlation methods and Fourier Transform-based correlation to improve speed

Fingerprint Matching

1. Correlation-based Matching Technique



Non-linear Distortion

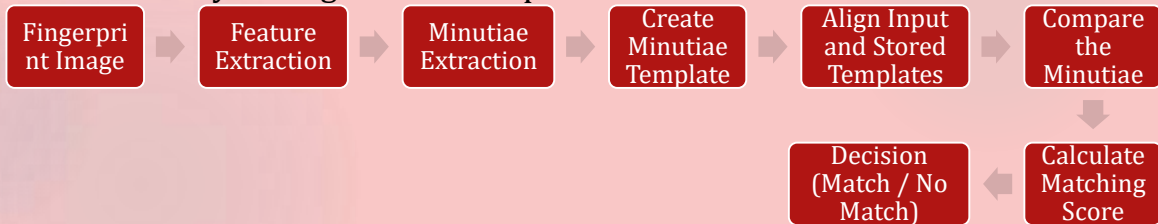


Variations in Image Quality

Fingerprint Matching

2. Minutiae-based Matching Technique

- ▶ The most widely used fingerprint matching technique.
- ▶ Compares the minutiae points (ridge endings and bifurcations) extracted from an input fingerprint with those stored in a database to determine whether they belong to the same person.



- ▶ Each minutia is represented as a triplet: $\mathbf{m} = (x, y, \theta)$ where
 - $x \rightarrow$ x-coordinate of the minutia,
 - $y \rightarrow$ y-coordinate of the minutia
 - $\theta \rightarrow$ orientation (angle) of the minutia

Fingerprint Matching

2. Minutiae-based Matching Technique

- ▶ Types of Minutiae Matching
 - ▶ 1. Global Minutiae Matching 2. Local Minutiae Matching
 - ▶ 3. Modern Approach (hybrid)
- ▶ **Global Minutiae Matching**
 - Compares the entire set of minutiae.
 - Requires alignment of the two fingerprints.
 - Uses translation, rotation, and scaling.
 - Provides high distinctiveness and accuracy.
 - A popular method for global matching is the Hough Transform.
 - It estimates the transformation parameters: $\Delta x, \Delta y, \theta$
 - After alignment, the minutiae are compared.



Fingerprint Matching

2. Minutiae-based Matching Technique

► Local Minutiae Matching

- Instead of comparing the entire fingerprint, this method compares small groups (local neighborhoods) of minutiae.
- Local structures remain almost unchanged.
- Invariant with respect to global transformation (e.g., translation, rotation, etc.)
- Therefore, global alignment is not required.
- Takes less time.



Fingerprint Matching

Global Minutiae Matching vs Local Minutiae Matching

Feature	Global Matching	Local Matching
Comparison	Entire fingerprint	Local minutiae neighborhoods
Difficulty	Simple algorithm	More complex algorithm
Alignment Required	Yes	No
Distortion Tolerance	Low	High
Computational Cost	Higher	Lower
Matching Accuracy	Higher	Moderate
Speed	Slower	Faster

► Modern Approach (hybrid)

- Most modern fingerprint recognition systems combine both global and local matching to achieve
 - ✓ High matching accuracy
 - ✓ Better distortion tolerance
 - ✓ Faster execution
 - ✓ Improved robustness

Fingerprint Matching

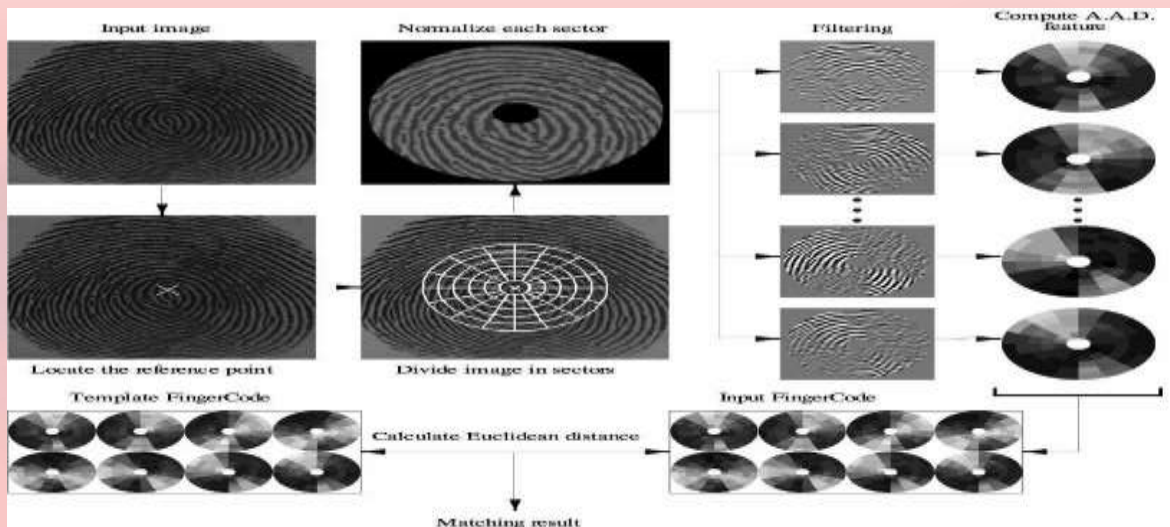
3. Ridge Feature-based Matching Technique

- ▶ Use fingerprint features other than minutiae, such as ridge orientation, ridge density, and spatial relationships between ridges.
- ▶ Purpose of using ridge features
 - Poor-quality fingerprints: Minutiae are difficult to extract reliably.
 - Time-consuming: Minutiae extraction requires more processing.
 - Improved accuracy: Ridge features can be combined with minutiae to improve accuracy and robustness.
- ▶ FingerCode → a local texture-based technique
 1. Locate the core/reference point of the fingerprint.
 2. Divide the fingerprint region into sectors around the core point.
 3. Extract local texture features from each sector using filtering.
 4. Combine the features from all sectors to form a FingerCode (feature vector).
 5. Compare the FingerCodes of two fingerprints using Euclidean distance.
 6. A smaller distance indicates greater similarity.

Fingerprint Matching

3. Ridge Feature-based Matching Technique

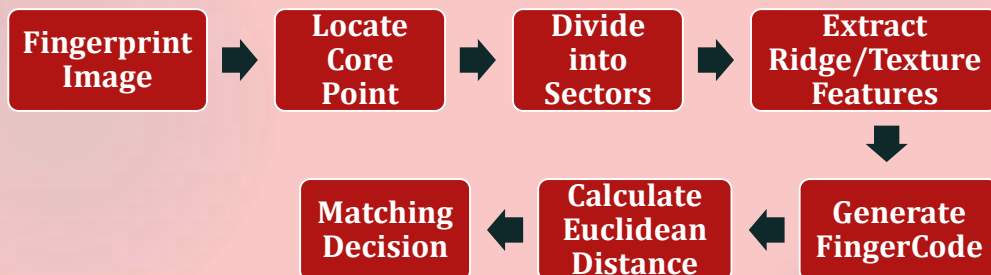
▶ FingerCode Techniques



Fingerprint Matching

3. Ridge Feature-based Matching Technique

- ▶ Other non-minutiae features that can be used along with minutiae:
 - ▶ Ridge orientation
 - ▶ Ridge density
 - ▶ Spatial relationship between ridge lines
 - ▶ Local texture information
- ▶ The process:



THANK YOU